

Performance & Strategy Benchmarks

Quantitative analysis of infrastructure efficacy.

Comparative Performance Matrix

Metric	Public API	Naive Self-Host	PolyAdapter
Latency (TTFT)	Variable (300ms+)	Stable (150ms)	Sub-100ms
Data Sovereignty	External (High Risk)	Internal (Secure)	Internal (Air-gapped)
Throughput	Rate Limited	Manual Scaling	Dynamic Batching
Unit Cost	\$4.00/1M	\$0.50/1M	\$0.02/1M

| Total Cost of Ownership (3Y)



Note: Figures based on 100M token/month utilization.

Strategic Implications

1. Sovereignty as a Service

Moving from public to private infrastructure is no longer just a security checkbox; it is a fundamental unit-cost reduction strategy.

2. Efficiency at Scale

Throughput is the ultimate lever. PolyAdapter's `PagedAttention` and `Continuous Batching` transform idle GPU capacity into active revenue generators.

3. Recommendation

Proceed with a 3-week pilot to validate the latency benchmarks against current production workloads.